



Indian Institute of Technology
Kanpur

Placement Brochure

2026-27

Department of Mathematics

Mathematics & Scientific Computing

BS · BS-MS (Dual Degree) · Double Major · M.Sc.



Students' Placement Office
Indian Institute of Technology Kanpur

WHY IIT KANPUR

INSPIRING EXCELLENCE AND INNOVATION SINCE 1959

ALSO SEE: IIT KANPUR OVERALL PLACEMENT BROCHURE



Established in 1959 and recognized as an Institute of National Importance, IIT Kanpur has built a distinguished legacy of excellence in science, engineering and interdisciplinary education. Its rigorous academic environment combines accomplished faculty, intellectually demanding curricula, advanced learning infrastructure and a culture that encourages analytical thinking, independent inquiry and cross-disciplinary exploration.

Research lies at the core of IIT Kanpur's academic ecosystem, driving advances across science, engineering and emerging technologies. With extensive laboratory and research infrastructure, the Institute fosters cutting-edge work at the interface of disciplines, translating fundamental research into impactful technologies and solutions for industry and society. IIT Kanpur's innovation ecosystem includes **1,000+ IPR filings**, **143 technologies licensed** and **232 industry partners**.

IIT Kanpur attracts an exceptionally accomplished student community through highly selective national-level admission processes across undergraduate, postgraduate and doctoral programmes. The resulting talent pool combines strong mathematical and analytical foundations with problem-solving ability, technical depth, intellectual curiosity and a demonstrated capacity to excel in demanding academic and professional environments.

IIT Kanpur has cultivated a strong interface between academia, industry and research, enabling knowledge and technology to move from the laboratory towards real-world applications. Collaborations with industry, government and academic institutions create opportunities for interdisciplinary research, technology development and engagement with complex problems of contemporary relevance. The Institute's research ecosystem is explicitly oriented towards innovation and technology development in interaction with universities, governments and industry.

#4 In NIRF Engineering 2025

#5 In NIRF Innovation 2025

7 Subjects in Global Top 100 QS World University Rankings

1,055 Acres of Lush Green Campus with World-Class facilities

MATHEMATICS IN THE GLOBAL TOP 100 QS WORLD UNIVERSITY RANKINGS BY SUBJECT 2026

#96 Mathematics

About The Department

The Department of Mathematics at IIT Kanpur was **established in 1960 with two faculty members**, as part of the Institute's founding academic structure. Over the decades, it has evolved into a **distinguished centre for mathematical education and research**, contributing to the advancement of mathematical sciences in India and internationally.

Today, the Department brings together a **vibrant academic community of 40+ faculty members and 250+ students**, supported by a **strong research culture with 500+ publications**. Its academic programmes combine **rigorous mathematical foundations with scientific computing, computational methods and interdisciplinary applications**, preparing students to tackle complex problems across science, technology and industry.

The Department offers programmes spanning **BS and BS-MS in Mathematics & Scientific Computing, Double Major and BSH/BSM pathways, M.Sc. Mathematics and Ph.D. programmes**. Designed to cultivate **mathematical maturity, computational capability and research aptitude**, these programmes provide students with a rigorous foundation for advanced study and high-impact professional careers.

Research at the Department spans a broad spectrum of pure and applied mathematics, scientific computing and interdisciplinary areas, including analysis, algebra, geometry, topology, differential equations, optimization, probability and statistics, numerical analysis, computational fluid dynamics, mathematical modelling and computational methods.

With a **strong foundation in mathematical theory and an expanding interface with computation, science and technology**, the Department develops graduates equipped for intellectually demanding roles across **research, technology, analytics, quantitative disciplines and industry**.



Prof. Akash Anand
HOD, Mathematics

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Dear Recruiters,

The Department of Mathematics shares the institute's vision of achieving excellence in teaching and research, and has always striven to succeed in this endeavour. Over the years, it has evolved into one of the premier departments in the country, providing excellent teaching and research in the Mathematical Sciences. In addition, research is carried out in interdisciplinary areas of science and technology, thereby contributing to the vibrant academic ambience of IIT Kanpur. We warmly welcome prospective employers to recruit from our accomplished graduates.

<https://iitk.ac.in/math/about>

About The Programmes

Undergraduate Programs (Mathematics & Scientific Computing)

BS / BS-MS (Dual Degree) / Double Major

The programmes combine rigorous mathematical foundations with scientific computing, computational methods and interdisciplinary flexibility. Admission to the BS programme is through JEE (Advanced), and students begin with a strong common core before progressing to advanced professional courses, departmental electives, open electives and undergraduate research projects. The **credit-based curriculum offers exceptional academic flexibility**, enabling students to pursue minors, second majors and interdisciplinary academic pathways alongside their primary programme. Students can also undertake undergraduate research through project courses, while the BS-MS pathway provides an extended route for advanced study and research. IIT Kanpur's flexible academic structure allows eligible students to complete the BS requirements in seven semesters and the BS-MS degree requirements in nine semesters. The programme's academic selectivity and growing appeal are reflected in its JEE (Advanced) profile: **in 2026, the BS Mathematics & Scientific Computing programme recorded an opening rank of 246 and a closing rank of 1,270 (OPEN category).**

The programme has also produced exceptional academic achievers, a Mathematics & Scientific Computing undergraduate has received the **President's Gold Medal for the best academic performance among graduating undergraduate students across the Institute**. The programme's breadth of mathematics, computation and interdisciplinary exposure prepares graduates for **research-intensive, quantitative and technology-driven careers**, while its flexible structure enables students to build domain expertise aligned with their academic and professional interests.

Postgraduate Program (Mathematics)

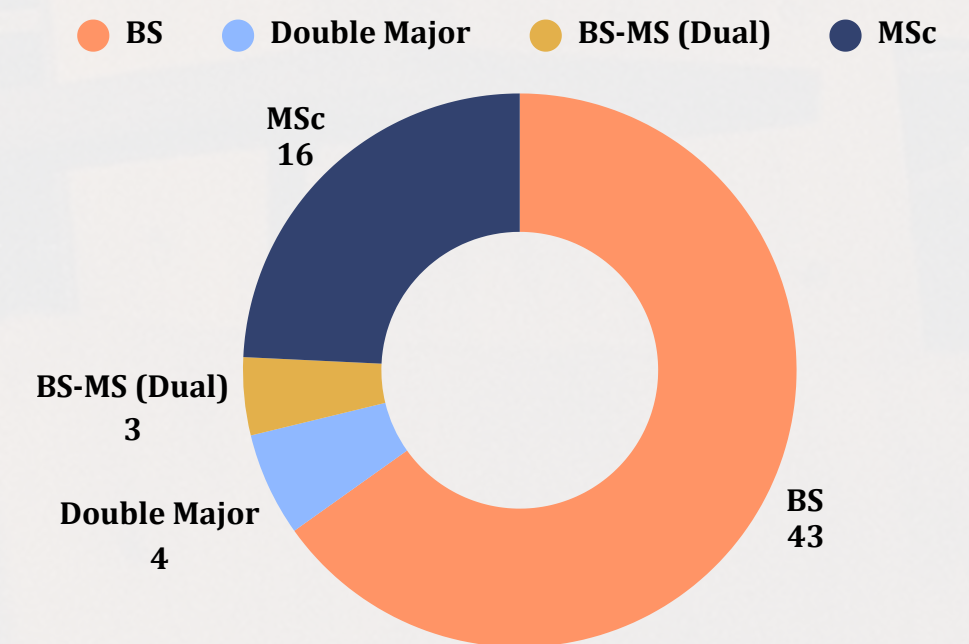
MSc

Admissions for this two-year postgraduate programme in Mathematics are made through the Joint Admission Test for MSc (JAM). The curriculum focuses heavily on logical reasoning, analytical thinking, and abstract problem-solving, along with practical training in mathematical computing, programming, and modelling. This blend prepares students for roles across data science, quantitative research, analytics, finance, software development, and Ed-Tech. While it provides a solid foundation for academic research, the programme's main strength lies in building industry-ready graduates capable of solving complex technical problems.

Demographics

PLACEMENT COHORT — 66 STUDENTS

Students participating in placements by programme



DISTINGUISHED FACULTY

DEPARTMENT OF MATHEMATICS • IIT KANPUR



ANAND A.
PhD: University of Minnesota

BAL, KAUSHIK
PhD: UPPA, France

BISWAS, ABHIJIT
PhD: Temple University

CHAVAN, SAMEER
PhD: University of Pune

GUIN, SATYAJIT
PhD: IMSc, Chennai

MALOO, ALOK KUMAR
PhD: TIFR, Mumbai

MUTHUKUMAR, T.
PhD: IMSc, Chennai

PATNAIK, S.
PhD: University of Cincinnati, USA

ROY, PROSENJIT
PhD: University of Zurich, Switzerland

SHARMA, SACHIN
PhD: IMSc, Chennai

SINGLA, POOJA
PhD: IMSc, Chennai

ARYA, VEDANSH
PhD: TIFR CAM

BANERJEE, MALAY
PhD: University of Calcutta

BISWAS, MRINMAY
PhD: IISER Kolkata

CHOWDHURY, INDRANIL
PhD: TIFR-CAM, Bengaluru

JHA, SOMNATH
PhD: TIFR, Mumbai

MANDAL, ASHIS
PhD: ISI, Kolkata

NADIMPALLI, SANTOSH
PhD: Université Paris-XI (Orsay)
and University of Leiden

PATTANAYAK, S. K.
PhD: CMI, Chennai

SAMUEL, PREENA
PhD: IMSc, Chennai

SHUNMUGARAJ, P.
PhD: IIT Bombay

THAKUR, AJAY SINGH
PhD: IMSc, Chennai

BAHUGUNA, D.
PhD: IIT Kanpur

BANERJEE, MOHUA
PhD: University of Calcutta

BONALA, NARASIMHA CHARY
PhD: Chennai Mathematical Institute

GANGULY, ARIJIT
PhD: TIFR Mumbai

KUBER, AMIT
PhD: University of Manchester, UK

MOHANTY, PARASAR
PhD: IIT Kanpur

NILAKANTAN, NANDINI
PhD: IISc, Bangalore

RATHISH KUMAR, B.V.
PhD: SSS IHL, Prasanthi Nilayam

SANKI, BIDYUT
PhD: IISc, Bangalore

SHEKHAR, SUDHANSHU
PhD: TIFR, Mumbai

DAR, APARNA
PhD: S.U.N.Y StonyBrook, USA
Emeritus Fellow

BAKSHI, KESHAB CHANDRA
PhD: IMSc., Chennai

BERA, SAYANI
PhD: IISc Bangalore

CHANDEL, VIKRAMJEET SINGH
PhD: IISc, Bengaluru

GHORAI, S.
PhD: Leeds, UK

KUMAR, ASHUTOSH
PhD: University of Wisconsin Madison

MUTHUKUMAR, P.
PhD: ISI, Chennai

PAL, A.
PhD: ISI, Kolkata

RAWAT, RAMA
PhD: ISI, Bangalore

SEN D.
PhD: ISI, Kolkata

SINGH, SAURABH KUMAR
PhD: TIFR, Mumbai

Explore Faculty & Research Profiles
→ iitk.ac.in/math/faculty



Courses Offered



Department of Mathematics • IIT Kanpur

Department Compulsory

- | | | |
|---|-----------------------------------|---|
| • Linear Algebra | • Real Analysis | • Data Structure & Algorithm |
| • Set Theory & Mathematical Logic | • Complex Analysis | • Probability & Statistics |
| • Abstract Algebra | • Ordinary Differential Equations | • Introduction to Electronics (UG only) |
| • Several Variable Calculus & Differential Geometry | • Partial Differential Equations | • Numerical Analysis & Scientific Computing I(& II) |

Department Electives

- | | | | |
|---|--|-------------------------------|---|
| • Computational Financial Mathematics | • Probability Theory | • Statistical Decision Theory | • ANN/ML Approach for Differential Equations |
| • Stochastic Calculus | • Econometrics | • Bayesian Analysis | • Linear Programming |
| • Markov Chain Monte Carlo | • Theory of Computation | • Inference | • Matrix Algebra and Linear Estimation |
| • Time Series Analysis: Forecasting and Control | • Statistical Computing | • Non-Parametric Inference | • Probabilistic Theory of Pattern Recognition |
| • Linear Regression and ANOVA | • Statistical & AI Techniques in Data Mining | • Mathematical Modelling | • Statistical Simulation, Data Analysis |
| • Game Theory | • Number Theory and Cryptography | • Techniques in Combinatorics | |
| | • Computability Theory | • Topology | |



Courses Offered



Open Electives

- | | | | | | |
|--|---|---|---|--|--|
| <ul style="list-style-type: none">• Design and Analysis of Algorithms• Operating Systems• Principles of Database Systems• Computer Networks• Computer Organisation• Theory of Computation• Computer Architecture• Deep Reinforcement Learning | <ul style="list-style-type: none">• Introduction to Machine Learning• Generative Artificial Intelligence• Probabilistic Machine Learning• Large Language Models (LLMs)• Deep Learning for Computer Vision• Randomised Algorithms• Parallel Algorithms• Data Mining | <ul style="list-style-type: none">• Advanced Algorithms• Big Data Visual Analytics• Natural Language Processing• Computer Vision and Image Processing• Algorithmic Game Theory• Theory of Multi-Armed Bandits• Parallel Computing• Parallel Programming• Principles of Programming Languages• Linux Kernel Programming | <ul style="list-style-type: none">• Topics in Distributed Systems• Indexing and Searching Techniques in Databases• Advances in DBMS• Programming for Performance• Sensing, Communication and Networks for smart wireless devices• Principles of Programming Languages• Computational Complexity | <ul style="list-style-type: none">• IoT System Design• Quantum Computing• Human AI Interaction• Functional Programming• Algorithms of Bayesian Networks and Causality• Machine Translation• Analysis of Concurrent Programs• Cryptography for Cybersecurity | <ul style="list-style-type: none">• Computer Systems Security• Modern Cryptology• Blockchain Queries• Optimization Techniques• Cryptographic Techniques for Privacy Preservation• Cyber security of critical infrastructure• Embedded & Cyber-Physical Systems |
|--|---|---|---|--|--|

Key Research Areas

DEPARTMENT OF MATHEMATICS • IIT KANPUR

-  Algebraic geometry
-  Commutative Algebra
-  Complex Analysis & Operator Theory
-  Computational Acoustics and Electromagnetics
-  Computational Fluid Dynamics
-  Differential Equations
-  Functional Analysis & Operator Theory
-  Harmonic Analysis
-  Homological Algebra
-  Image Processing
-  Mathematical Biology
-  Number Theory & Arithmetic Geometry
-  Numerical Analysis and Scientific Computing
-  Operator Algebra
-  Representation Theory
-  Set Theory and Logic
-  Several Complex Variables
-  Topology and Geometry
-  Tribology



Major Achievements



Students of our Department actively participate in leading national and international competitions, earning distinctions in programming, quant, product design, and communications research.

- **ICPC Asia West Finalist'26:** Represented our institute at the prestigious Asia-West regional round of the world's largest programming contest and was finalist.
- **Inter IIT Tech. Meet 14, 2025:** IIT Kanpur secured Overall 2nd Position at Inter IIT Tech Meet 14.0, standing tall among all IITs with consistent excellence across 7 problem statements.
- **Exceptional Performances in ICPC, IICPC Codefest, Quantfest and several more.**
- **Selection in TIFR's Visiting Students Research Program (VSRP).**
- **Optiver FututreFocus Programme 2025:** Selected among top applicants across India for Optiver Future Focus Mumbai 2025 for Quantitative training.
- **Expert and Candidate Master Title-Holders on Codeforces.**
- **Google Girl Hackathon 2025:** Advanced to the semi-finals among thousands of participants across India.

Technical & Analytical Skills

► Software Engineering & Systems Design —

Data structures & algorithms, object-oriented design, distributed systems fundamentals, REST/gRPC APIs, database design (SQL/NoSQL), and writing scalable, production-quality code in C++/Java/Python/Go.

► Quantitative Finance & Stochastic Modeling —

Stochastic calculus, Itô processes, Monte Carlo simulation, derivatives pricing (Black-Scholes, binomial trees), and portfolio risk analytics (VaR, CVaR). High-frequency data analysis, volatility modeling (ARCH/GARCH), cointegration-based pairs trading, statistical arbitrage strategies, and dynamic trend forecasting.

► Algorithmic & Low-Latency Engineering —

High-performance computing, memory-efficient C++, multithreading and parallel programming, data structure design, and algorithmic complexity optimization.

► Data Modeling & Machine Learning —

Bayesian inference (MCMC), supervised/unsupervised learning pipelines, deep learning and reinforcement learning, and convex/stochastic optimization at scale.

► Data Engineering & Analytics at Scale —

SQL/NoSQL data pipelines, big-data processing frameworks, exploratory data analysis, and dashboarding for business decision-making.



Past Recruiters



Past Recruiters

NOMURA

Morgan Stanley

 **Money Forward**

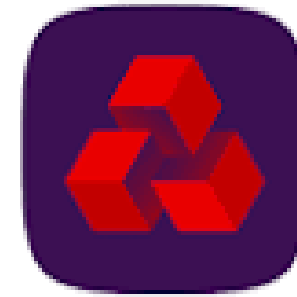
 **Razorpay**  **Groww**

RISA

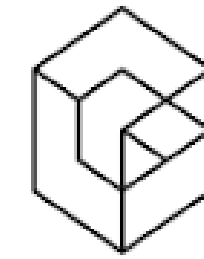
 **ENPHASE**

 **Capital One**

 **Expedia**



NatWest



genpact

and many more ...

Goldman Sachs

WELLS FARGO

 **standard chartered**

AMERICAN EXPRESS

BlackRock



Deutsche Bank

 **BARCLAYS**

JPMorganChase

BCG **BOSTON CONSULTING GROUP**

McKinsey & Company

citibank

BAIN & COMPANY 

paytm

 **BLUESTONE**

 **BNY**

Some Edtechs Profiles-

ALLEN

 **Sri Chaitanya Techno School**



 **Aakash**

 **MOTION ACADEMY**

 **NINE EDUCATION IIT ACADEMY**

 **NARAYANA EDUCATIONAL INSTITUTIONS**

 **PHYSICS WALLAH**

 **BT BAKLIWAL TUTORIALS-IIT**

RECRUITMENT PROCEDURE

ALSO SEE IIT KANPUR'S PLACEMENT BROCHURE FOR PLACEMENT POLICIES

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- ➔ The Students' Placement Office appoints a Point of Contact (PoC) for your organization.
 - ➔ You will need to contact your PoC or any Placement Coordinators (or email spo@iitk.ac.in) to obtain your account credentials for the Recruitment Automation System (RAS).
 - ➔ You need to fill out a Job Announcement Form (JAF) or Internship Proforma (IP) on the portal to share details of the offered profile.
 - ➔ After verification of the submitted details by the PoC, the job/internship openings become visible on the portal as per the schedule decided by the Placement Team.
 - ➔ Students submit their resumes on the RAS before the application deadline specified for the company.
 - ➔ The company conducts its Pre-Placement Talk (PPT) and tests/screening process after finalizing the schedule with the Placement Team.
 - ➔ The company submits the final list of shortlisted students for the interview process.
 - ➔ The Placement Office allots interview dates and assigns a PoC for the interview process.
 - ➔ The company confirms the interview dates with the Placement Office via email.
 - ➔ At the end of the interview slot, the company provides the list of selected and waitlisted students.
 - ➔ The Placement Office notifies the company regarding the acceptance of selected students.
 - ➔ The company sends offer letters to the selected candidates in accordance with the details specified in the JAF/IP.

PLACEMENT SEASON 2026-27

TIMELINE

September 2026 ONWARDS

- Registrations for Companies for the Placement Drive

OCTOBER 2026 (FIRST WEEK)

- Pre-Process Activities Begin
- Online Test and Shortlisting
- Pre-Placement Talks Start

DECEMBER 2026

- Interviews Start for the Placement Drive

15TH Jan 2027 ONWARDS

- Placement Drive Phase 2 Starts

Interested in Hiring ? Contact Us

For Undergraduate Talent
Contact



**Faculty Placement
Coordinator**

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Coordinator**

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**Department Placement
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INDIAN INSTITUTE OF TECHNOLOGY KANPUR



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